Network ATP in Azure Local

Agenda

- **Objective:** Explore Network ATC in Azure Local
- **Scope:** Network ATC architecture in Azure Local, including intents, intent types, various intent-based network topologies, overrides, and their management by using PowerShell Network ATC cmdlets

Network ATC

- **Network ATC** provides **intent-based automation** for host networking
- **Offers a range of benefits:**
  - Reduced deployment time and configuration complexity
  - Minimized risk of human error
  - Symmetry, consistency, and adherence to best practices
  - Configuration drift correction
  - Automatic storage adapter configuration, including VLAN + IP assignment for storage
  - Automatic cluster network naming
  - Windows Admin Center integration
  - Proxy configuration management
  - Live Migration tuning
  - Scope detection for Network ATC PowerShell cmdlets

What Network ATC Is and Is Not

- **Network ATC** = full product name
  - Not an acronym
- **Designed to provide:**
  - Automation, validation, consistency
  - Microsoft-validated best-practice enforcement
- **Not intended to:**
  - Replace physical switch configuration
  - Manage unsupported NICs or asymmetric layouts

Terminology: Intent

- **Intent** = the definition of how physical adapters will be used
- Each intent includes:
  - One or more **physical adapters**
  - One or more **intent types**
- A physical adapter may belong to **only one intent**

Terminology: Intent Types

- **Management**: node access; max 1 intent per cluster
- **Compute**: VM traffic; unlimited per cluster
- **Storage**: SMB + S2D; max 1 intent per cluster
- **Stretch**: storage-like but **no RDMA**; max 1 intent per cluster
- Any combination allowed per intent

Terminology: Overrides

- Overrides allow customization of:
  - VLANs
  - DCB/QoS
  - Adapter properties
  - Storage IP generation
- OS limitations still apply (e.g., cannot modify SR-IOV on existing vSwitch)

Network Symmetry Essentials

- NICs must match:
  - Make
  - Model
  - Speed
  - Configuration
- Symmetry enforced per node and across cluster
- Required for SET and RDMA

Automatic IP Addressing for Storage

- Storage intent triggers:
  - NIC property configuration
  - DCB configuration
  - VLAN assignment
  - vSwitch creation (if needed)
  - Automatic IP address assignment
- Works for **switched** and **switchless** topologies

Cluster Network Naming

- Replaces generic names like “Cluster Network 1”
- Names reflect actual purpose:
  - management
  - storage_711, storage_712, etc.
- Improves diagnostics and troubleshooting

Live Migration Optimization

- ATC manages:
  - Max concurrent LM operations
  - LM network selection
  - Transport selection (SMB / Compression / TCP)
  - RDMA bandwidth budgeting
- Defaults align with OS recommendations

Proxy Configuration via ATC

- Cluster-wide proxy settings managed centrally
- Supports:
  - ProxyServer
  - ProxyBypass
  - AutoDetect
  - AutoConfigURL
- Applies to all nodes, including future ones

Scope Detection

- ATC automatically identifies the cluster context
- No need to specify `-ClusterName` when running commands on a node
- Simplifies cmdlet usage and reduces errors

Modifying VLANs After Deployment

- Use `Set-NetIntent` to change VLANs
- Works for:
  - ManagementVLAN
  - StorageVLANs
- Safe & supported method to adjust post-deployment

Prerequisites for Network ATC

- Hardware must be certified
- NICs must be symmetric
- NIC names must be identical cluster-wide
- Required Windows features:
  - Network ATC
  - Hyper-V
  - Failover Clustering
  - Data Center Bridging
  - FS-SMBBW

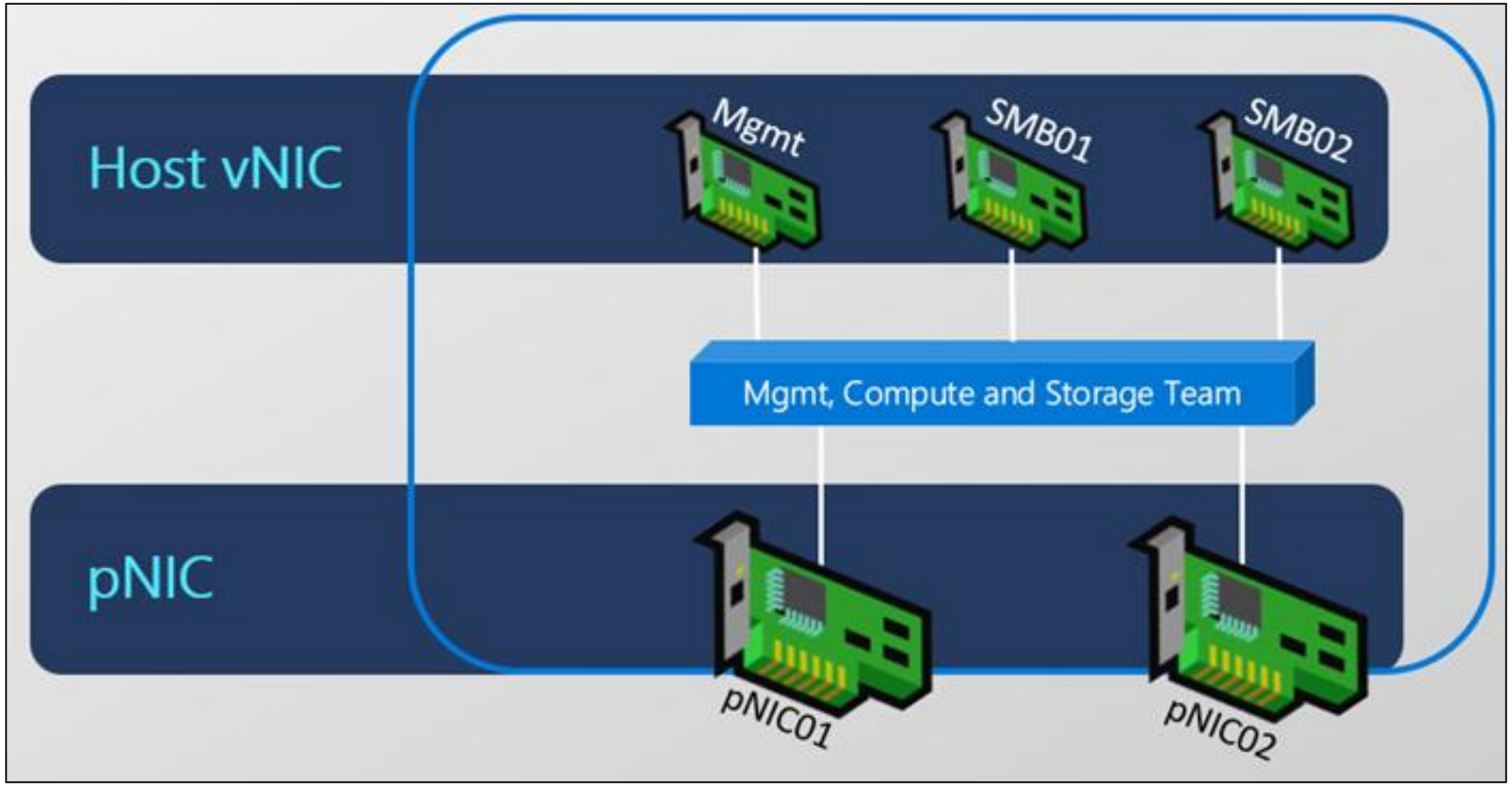
Common ATC Cmdlets

- `Add-NetIntent`
- `Set-NetIntent`
- `Remove-NetIntent`
- `Get-NetIntentStatus`
- `Update-NetIntentAdapter`
- Override creation cmdlets
- Global override cmdlets

Fully Converged

- One intent with:
  - Management
  - Compute
  - Storage
- Applied to the same NIC team

Fully Converged

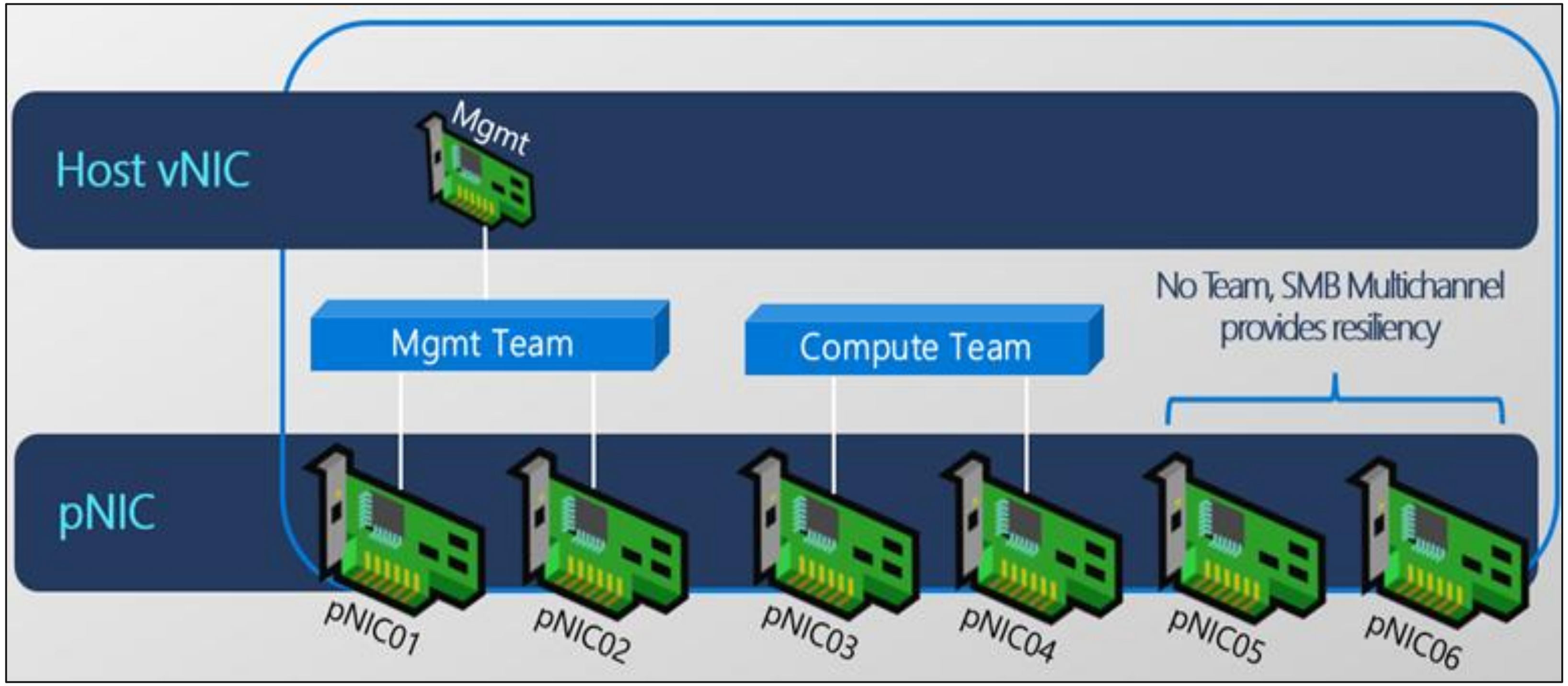



```
Add-NetIntent -Name ConvergedIntent
              -Management -Compute -Storage
              -AdapterName pNIC01, pNIC02
```

Fully Disaggregated

- Separate intents:
  - Management
  - Compute
  - Storage
- Each mapped to different NIC pairs
- Highly modular layout

Fully Disaggregated

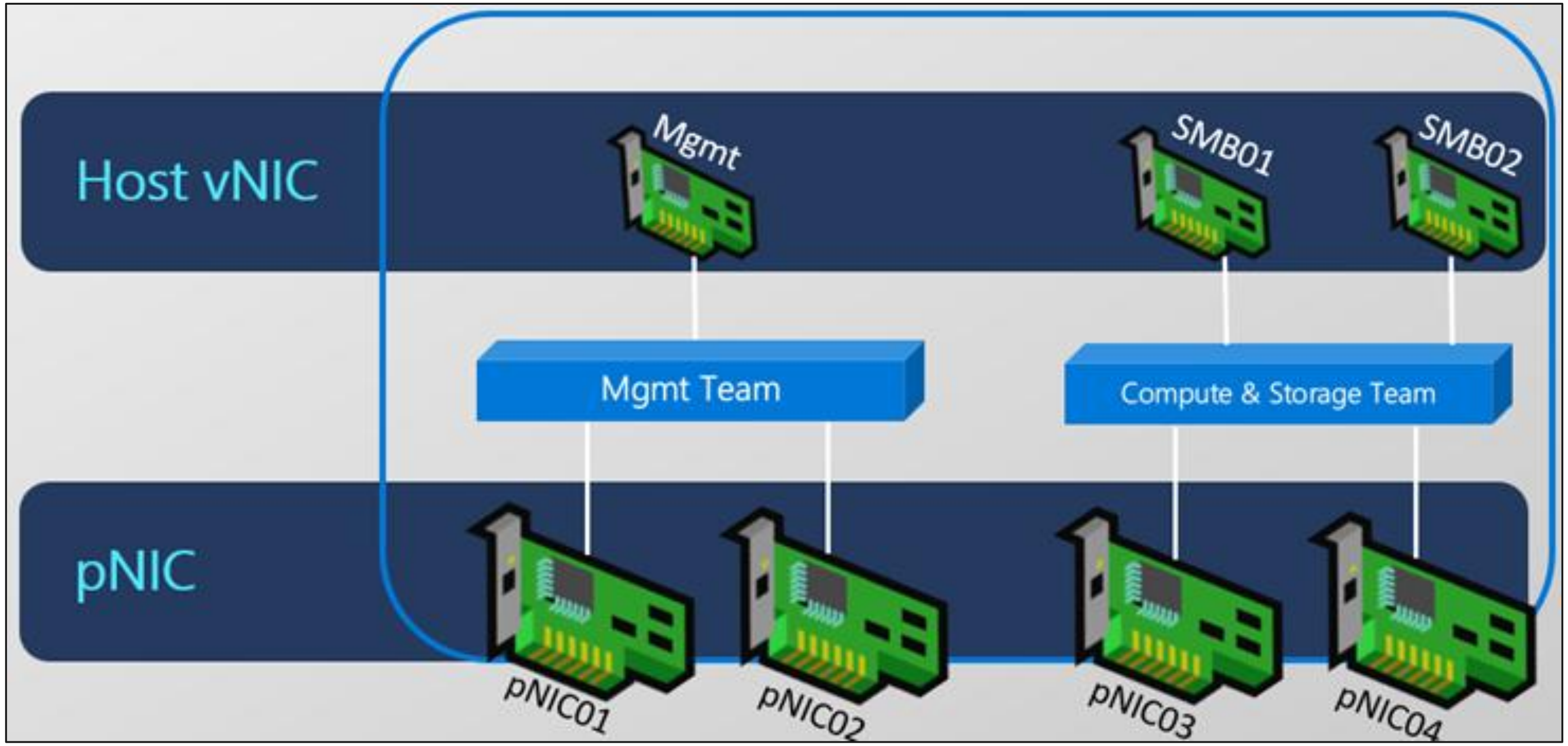



```
Add-NetIntent -Name Mgmt -Management -AdapterName pNIC01, pNIC02
Add-NetIntent -Name Compute -Compute -AdapterName pNIC03, pNIC04
Add-NetIntent -Name Storage -Storage -AdapterName pNIC05, pNIC06
```

Converged Compute and Storage, Separate Management

- Two intents deployed cluster-wide
  - Management on pNIC01, pNIC02
  - Compute + Storage combined on pNIC03, pNIC04
- Clear separation between infra access and workload & storage traffic
- Enables simplified NIC topology while maintaining functional isolation

Converged Compute and Storage, Separate Management

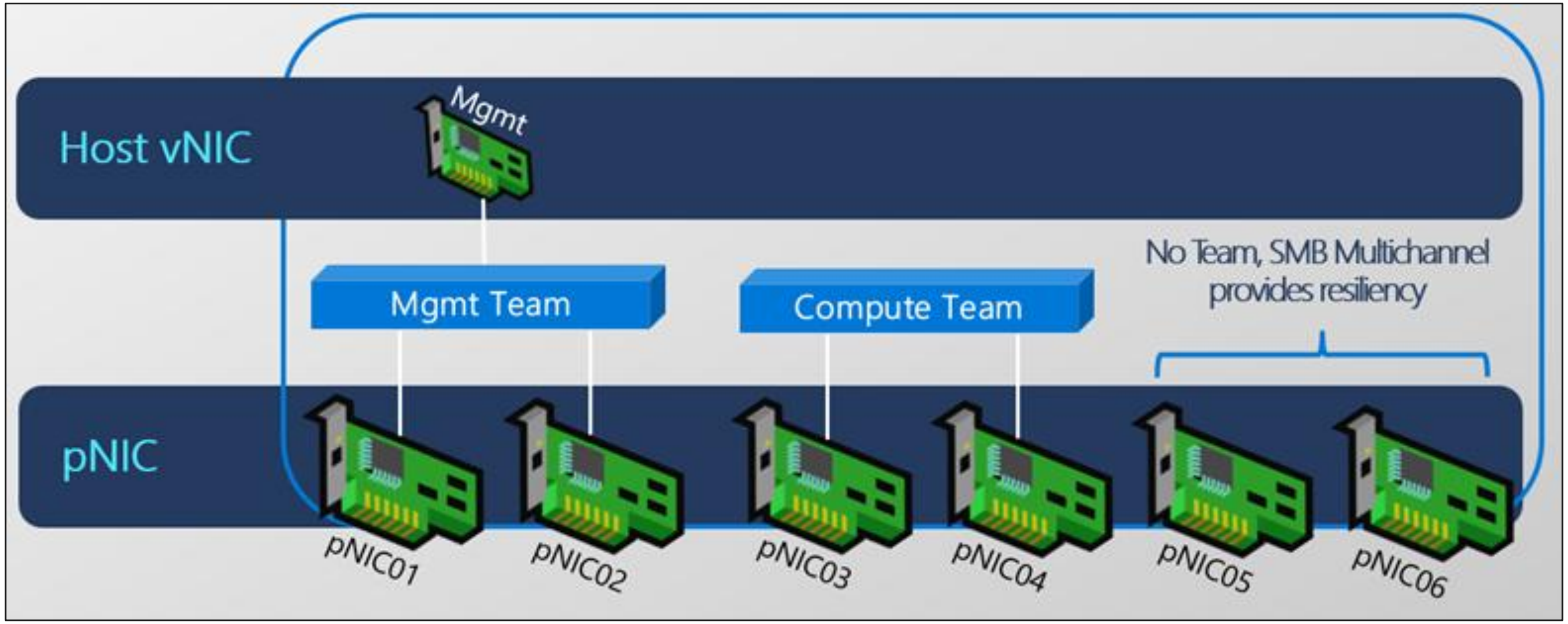



```
Add-NetIntent -Name Mgmt -Management -AdapterName pNIC01, pNIC02
Add-NetIntent -Name Compute_Storage -Compute -Storage -AdapterName pNIC03, pNIC04
```

Storage-Only

- Only storage traffic is managed by Network ATC
- Management and Compute adapters are left unmanaged
- For environments with manually configured management/compute networks
- ATC configures storage NICs, VLANs, DCB, and RDMA

Storage-Only

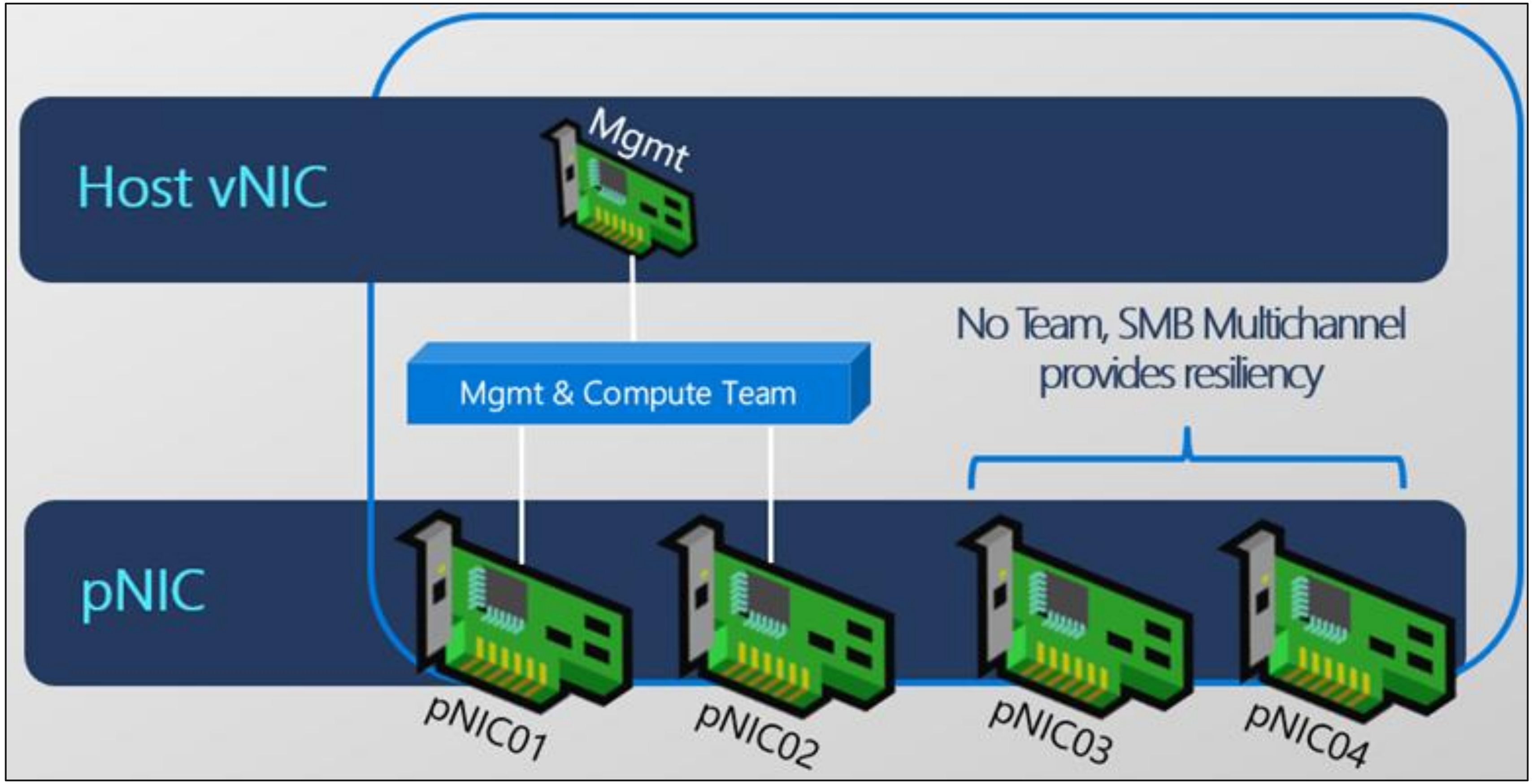



```
Add-NetIntent -Name Storage -Storage -AdapterName pNIC05, pNIC06
```

Compute and Management

- Compute and Management combined within one intent
- Storage excluded (for SAN-based or manually configured environments)
- Reduces number of intents while maintaining consistent configuration
- Ideal where storage fabric is independent of host NICs

Compute and Management

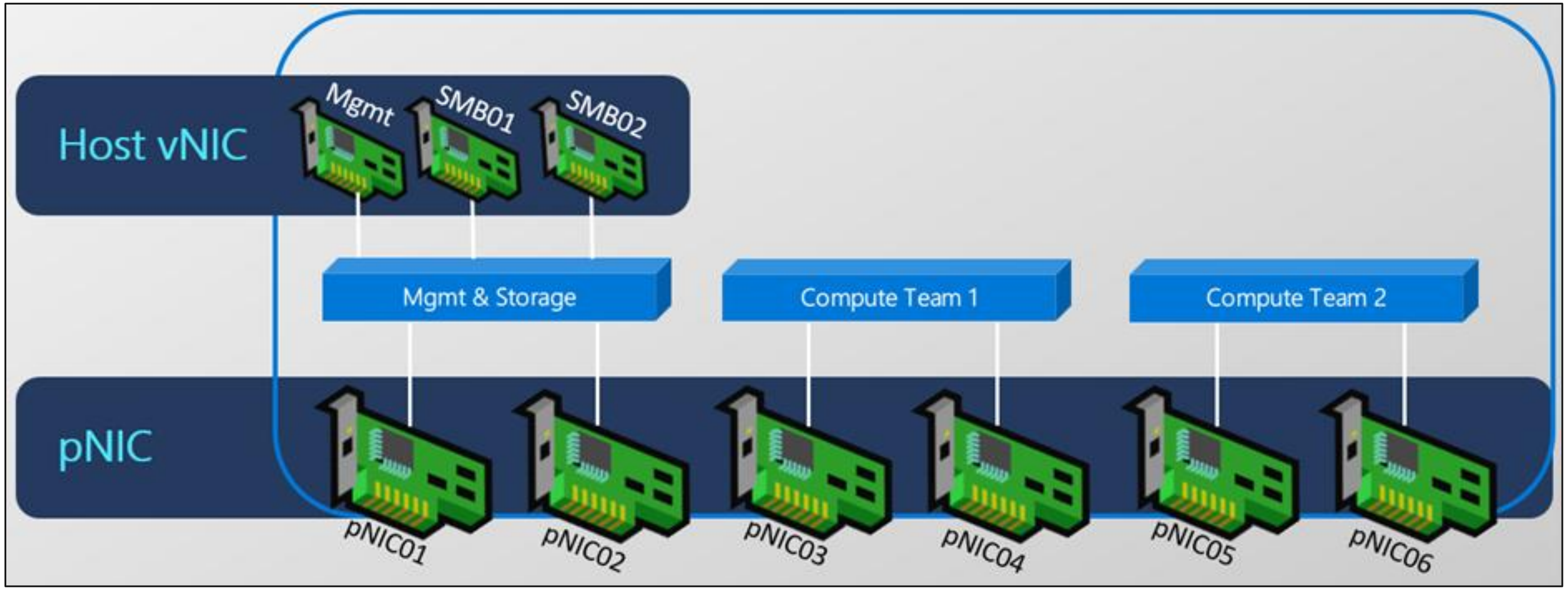



```
Add-NetIntent -Name Management_Compute
              -Management -Compute
              -AdapterName pNIC01, pNIC02
```

Multiple Compute (Switch)

- Supports multiple compute intents, each tied to different NIC pairs
- Enables multi-fabric VM networking
- Each compute intent creates its own vSwitch
- Useful in scenarios requiring tenant isolation, traffic engineering, or multi-network VM placement

Multiple Compute (Switch)




```
Add-NetIntent -Name Compute1 -Compute -AdapterName pNIC03, pNIC04
Add-NetIntent -Name Compute2 -Compute -AdapterName pNIC05, pNIC06
```

Default VLAN Assignments

- Storage VLAN defaults:
  - 711, 712, 713, 714, …
- Auto-assigned per adapter order
- Overrides allowed via StorageVLANs parameter

Automatic Storage IP Addressing

- **Default schema:**
  - Adapter 1 → 10.71.1.x
  - Adapter 2 → 10.71.2.x
  - Adapter 3 → 10.71.3.x
- **Ensures:**
  - Uniformity
  - Collision avoidance
  - Switchless support

Cluster Network Settings (Defaults)

Network ATC configures cluster network features:

```
EnableNetworkNaming = $true
EnableLiveMigrationNetworkSelection = $true
EnableVirtualMachineMigrationPerformance = $true
VirtualMachineMigrationPerformanceOption = Auto-calculated
MaximumVirtualMachineMigrations = 1
MaximumSMBMigrationBandwidthInGbps = Auto
```

Default Data Center Bridging (DCB) Configuration

ATC applies Microsoft-validated DCB policies:

- Cluster heartbeat
  - Priority 7, 2% or 1% bandwidth
- SMB_Direct (RDMA)
  - Priority 3, 50% bandwidth
- Default traffic
  - Priority 0, remaining bandwidth

Storage VLAN Defaults

Storage intents use default VLAN assignments:

- Adapter 1 → 711
- Adapter 2 → 712
- Adapter 3 → 713
- Adapter 4 → 714
- Additional up to 719

Storage Automatic IP Schema

IP addressing template:

- pNIC1 → 10.71.1.X / VLAN 711
- pNIC2 → 10.71.2.X / VLAN 712
- pNIC3 → 10.71.3.X / VLAN 713

Overrides: Storage IP Customization

Use overrides when custom ranges are required:

- `New-NetIntentStorageOverrides`
- Disable auto IP generation
- Pass override to `Add-NetIntent`

Global Overrides: Cluster Features

ATC supports **GlobalClusterOverrides** to tune:

- Naming behavior
- Live Migration network selection
- Performance transport
- Bandwidth controls
- Cluster-wide network behavior

Global Overrides: Proxy Settings

Configure once, apply everywhere:

- ProxyServer
- ProxyBypass
- AutoDetect
- AutoConfigURL

Applying Proxy Override (Example)

```
# Define the proxy server parameters and local bypass domains
$ProxyParams = @{
    ProxyServer = "http://contoso.com"
    ProxyBypass = "localhost;127.0.0.1;*.contoso.com;<local>"
}

# Generate the global proxy override object
$ProxyOverride = New-NetIntentGlobalProxyOverrides @ProxyParams

# Apply the proxy configuration object globally across the cluster boundary
Set-NetIntent -GlobalProxyOverride $ProxyOverride

# Retrieve the global configuration to verify implementation
(Get-NetIntent -GlobalOverride).ProxyOverride
```

Managing ATC After Deployment

Core management tasks:

- Add/remove cluster nodes
- Monitor `Get-NetIntentStatus`
- Update NIC membership
- Apply overrides
- Re-apply configurations using `Set-NetIntentRetryState`

Adding a Cluster Node

Steps:

- Install required features
- NICs must match naming and symmetry
- Add node
- ATC automatically applies existing intents to the new node

Updating NIC Membership in an Intent

If NICs change:

- Validate NIC status (`Get-NetAdapter`)
- Use `Update-NetIntentAdapter`
- Confirm via `Get-NetIntent`

Removing an Intent

Removing intent does not remove:

- vSwitches
- QoS objects
- Flow control settings

Explicit cleanup is required:

- Remove all intents
- Remove vSwitches
- Remove NetQoS objects
- Remove flow control settings
- Reset CIM sessions

Common Error: AdapterBindingConflict

Occurs when:

- NIC is bound to incompatible vSwitch
- NIC has conflicting components enabled

Fix:

- Remove vSwitch or unbind component
- Run `Set-NetIntentRetryState`

Common Error: RDMANotOperational

Caused by:

- Built-in NIC drivers
- SR-IOV disabled
- RDMA disabled

Fix:

- Update NIC driver
- Enable SR-IOV
- Enable RDMA in BIOS

Automatic Remediation (Drift Correction)

ATC monitors for drift and remediates:

- MTU changes
- QoS changes
- Adapter properties
- VLAN changes
- Live Migration settings

opsgility